2.11.

Listing 2.11 Top k elements of a list

```
function topK(A, k)
     heap ←
DWayHeap()
   #1
     for el in A
do
#2
        if (heap.size == k and heap.peak() < el)
then                          #3

heap.top()
   #4
        if (heap.size < k) then
                                                #5

heap.insert(el)
   #6
     return heap
                  #7
```

#1 Create an empty min-heap
#2 Iterate through the elements in the array `A`
#3 If we have already added at least `k` elements, check if current element is larger than the top of the heap
#4 In that case, we can safely remove and discard the heap's top, as it won't be among the `k` largest elements. After this the heap will have `k-1` elements
#5 If, at this check, the heap size is less than `k`...
#6 ... we must add current element